

Challenging the Standards of Mental Care: An Analysis of Self-Rating with respect to Sensor based State Detection

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ABSTRACT

Despite all the developments in AI and Ubiquitous computing, management of mental health and mental disorders mainly relies on human assessments, like daily self-reporting and standard psychological questionnaires. Self-reporting however, in addition to compliance issues, comes with the drawback of being subjective and thus often inaccurate. In a depressive episode, it is hard to recall the manic phase of the last weeks. Thus, mainly experienced and self-aware patients can use self-reports effectively. Even though, in the last 15 years, sensor-based objective algorithms to monitor mental disorders showed promising results [3],[1], such systems are not established in psychiatric care. We believe it would help psychiatrists and patients greatly to consider using objective sensor-based support systems if it would be possible to visualize the drawbacks of self-reporting and the abilities of sensor-based analysis. Our work provides a direct comparison of the performance of sensor-based analysis, self-reporting, and psychological diagnosis. It is based on a real-life data set collected with psychiatric patients. It consists of smartphone sensor data, respective daily self-reporting questionnaires, and a ground truth of standardized psychiatric scale tests. As a highlight of this work, in the evaluation, we can provide evidence that observations of deferred self-perception of patients concerning their mental states, as doctors reported to us, can be measured in the sensor data.

CCS CONCEPTS

• **Human-centered computing** → *Ubiquitous and mobile devices.*

KEYWORDS

management of mental health, deferred self-perception, sensor-based state recognition

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1 INTRODUCTION

Modern medicine can draw upon various technological applications to support diagnosis and treatment, and robots and endoscopic cameras allow minimally invasive surgery with the surgeon not even in the same room. Despite all these developments, until now psychiatric care and management of mental disorders, barely rely on technology neither for diagnosis nor treatment. On the contrary psychiatrists and psychotherapists still mainly depend on the self-ratings of patients and reports of relatives. Yet, these self-ratings will often be biased and dependent on the subjects' abilities to perceive themselves correctly. Psychiatrists themselves reported to us though that they experience many patients taking up to a week to notice a change (be it improvement or decline) in their mental state. To our knowledge, this delay in self-perception has never been measured objectively or in detail. Still, psychiatrists understand that this can mean they get to see the patient only after a change has manifested already instead of being able to adjust treatment immediately. However, so far they have not been presented with a better solution, they would be willing to trust.

Considering the achievements of pervasive and ubiquitous computing in various fields including health care, activity recognition recommends itself as an application for mental care, since affective disorders, like bipolar and depressive disorders, are expressed by changes in activity and behavior. A number of research to provide mechanisms for objective sensor-based analysis of behavior (e.g. [8]) have been published in recent years.

Nevertheless, such systems are still not established in psychiatric care. This is, because many doctors and patients find it hard to trust the reliability of a technological system, specifically in regards to a complex aspect such as the human mind, and prefer to stick with what they know [4]. One of the reasons for this is that so far little effort has been put into providing an understandable way to visualize the results of sensor-based methods as well as showing the limitations of current standard methodologies (mainly self-reporting), so even laypeople can grasp the potential of digital and sensor-based methods compared to their known standards.

1.1 Objectives and Contribution

In this work, we wanted to take up this challenge and provide visualization and evaluation of the commonly used methods as self-reporting of affective disorder patients against the psychiatric diagnosis and sensor-based state detection.

We start our report by introducing the different data modalities that were available. Then in the first evaluation we provide both, a visual qualitative comparison of the data modalities (namely how

the sensors compare to self-reporting and the diagnosis visually), but also a quantitative evaluation of calculated similarities between the different modalities.

In the second part, we go into more depth and analyze self-assessments in comparison to the actual diagnosis. This allows us to display deferral of self-perception - as mentioned above - by comparing the self-reporting to the diagnosis by step-wise shifting them against each other for several days and calculating the similarity of the respective data curves after each shift. We end this paper with a discussion of the results of the potential and limitations of this work.

1.2 Related Work

In the area of mental healthcare, the majority of systems deployed to date focus on supporting self-monitoring. [11], [12], [5] and [13] or [7], just to name a few studies using self-rating questionnaires. Even though these studies evaluate the quality of self-rating, they do not go into the known drawbacks of bias. Furthermore, these systems often require constant interaction and feedback from the patient, which often affects adherence in a negative way. Sensor-based Telesystems to help recognize the state and support the management of mental disorders have been introduced in the last few years. From our own work [3], to more recent [6] and [9], many approaches have been trying to develop such systems. However, as [4] have concluded, for many digital support systems the complexity of the system and the lack of trust in the method are barriers for both doctors and patients to engage with them.

2 COLLECTION OF DIFFERENT DATA SOURCES

During a study (approved by the ethics board of the local university clinic) deployed in a rural-area psychiatric hospital with patients diagnosed with bipolar affective disorder, an extensive data set was recorded. (See [2]), and [3]). The study was conducted with a total of 10 patients. After a detailed introduction and signing the informed consent, the participants (patients) received a commercially available smartphone with the specifically developed App installed, if they did not own a smartphone or the required App was installed on their own smartphone. The App was designed to collect behavioral patterns by the phone's internal sensors but also to record a daily self-reporting questionnaire. Each patient was supposed to participate at least for 12 weeks (3 months) since previous calculations had concluded that a typical bipolar patient would undergo at least one change in state within 3 months. The study was designed as a real-life study, therefore some data issues occurred, meaning that patients switched off their phones for some days, so some patients collected less data. Others, on the other hand, participated even longer than the 12 weeks. In summary, we collected data between 50 and 140 days per patient. In total, these are more than 800 days of sensor data documenting a total of 24 state changes (16 improvements of state, 8 declines of state, 8 extrema - meaning change of direction e.g. a person improving state and then declining again). Next to sensor data collection and self-reporting also frequent psychiatric assessments were performed.

Three different data modalities were gathered:

2.1 Psychological and Psychiatric Assessment:

To validate the sensor data, and to provide ground truth, psychological state examinations (psychological standard scale tests, e.g. HAMD, Young Mania, etc.) were performed every three weeks. Performing those standard tests more frequently was not advised since it would have easily resulted in "learning" the tests and thus biased their outcome. To get more data points we asked psychologists to interview the patients via phone calls in between every three weeks and rate their impressions in the provided scale. These examinations resulted in grades for each measurement on a 7-point HAMD scale between -3 for severe depression and +3 for severe mania with intermediate steps of depression, slight depression, normal (0) slight mania, and mania (half grades were possible). These examination data points are called "ground truth" or "diagnosis" in further text. To make the ground truth easily comparable to the other data modalities, the 7-point-scale values were normalized to a 1-5 scale (see also the next paragraph "Self-Assessment").

2.2 Self-Assessment:

During the Study, the patients were asked to fill out a simple daily self-assessment questionnaire. In these, the patients were asked to rate their day. The questions included grading the amount of performed activities and places visited, rating their psychological, physical, and activity level as well as answering specific questions about depressive and manic state. The questions about manic and depressive states were derived from clinical questionnaires related to bipolar disorder episodes and included questions about the quality of sleep, meals taken, places visited, and the duration of these activities. The scale of the questionnaire was kept in a 1-5 range. The study took place in Austria, so we used the 1-5 similar to inverted school grades (1 - very bad, 5 excellent) because this scale was presumed to be natural to the patients and easier to handle.

2.3 Smartphone Sensor Data:

As already mentioned, the smartphone app was designed to collect raw data from the phone's internal sensors. These sensors included acceleration, GPS traces (anonymized), phone-call behavior (length of calls, etc.), and sound (voice frequency, etc.). All data was anonymized and for Sound only randomly shuffled snippets of sound were stored to guarantee the patients' privacy. The recorded data included a total of more than 800 days of sensor recordings (an average of 600 traces per sensing modality).

Using this dataset successfully implemented the detection of patients' state changes and recognition of mental state out of objective sensor data. With the acceleration and location features recorded by the smartphone of the patients, supervised learning standard pattern recognition techniques were applied to the data to identify which state a patient had been in. The classes for the classification were defined according to the diagnosis provided by psychologists (depressive, normal, manic with different degrees - up to 7 classes possible). With a Naive Bayes classifier, the classification was evaluated using a 500 times cross-validation and reached an accuracy of 70-80%. (Table 1)

Also, a State Change Detection was implemented, which only relies on a model of a single "default state". All points falling outside this model are classified as a "change". To determine the border

between in the model and outside the model (threshold), a set of values was tested, resulting in a precision/recall of more than 97%. Table 1 Details about state recognition and state change detection have been published in: [3] and [2]].

To make these results comparable with other data modalities, the results of the state recognition in the different sensor modalities (acceleration and location) were averaged and normalized to the inverted 1-5 scale.

3 QUALITATIVE AND QUANTITATIVE COMPARISON OF THE DATA MODALITIES

Figure 1 displays a visual comparison of the three data modalities: diagnosed scores (Ground Truth), self-rating (SelfAss), and the normalized recognized state (Sensor-Results) for four different patients.

At first glance, it is visible, that all 3 modalities match quite well for P3. For P5 (top in Figure 1) they match somehow - specifically sensor results and ground truth match well, but self-rating not entirely. For P6 (lower left in Figure 1) it seems like the self-reporting is an inverted form of the sensor results, while sensors and ground truth have some similarity in shape. For P7 (lower left in Figure 1) nevertheless, it seems like self-reporting and sensors match to some extent in shape, but ground truth does not so much.

Also visible at the second glance is that sensor data displays changes in the state that are missed by the ground truth. For example, after an increase, the state of p3 breaks down around day 15. In the diagnosed scores this gets only recognized around day 40. Maybe the doctor's appointment in-between was missed, maybe the doctor deemed the appointment not necessary since the patient was doing well. At this point (displayed in all three modalities) the condition of the patient has dropped to a very bad state. The sensor data on the other hand clearly shows the breakdown immediately - with the self-reporting lagging behind a couple of days. Had the sensor system had the ability to send out a warning when the breakdown started, the doctor could have reacted right away, and the extreme breakdown might have been avoided! But also for the other patients changes in state are visible in the sensor data that are not captured by the diagnosed ground truth. For the other patients, this effect is not as devastating as for P3. However, P5 also shows some potential for sensor-based state monitoring because the sensor data show some clear fluctuations in a time frame where no diagnostics happened.

For a quantitative comparison, we calculated the cumulative distance between sensor results and the diagnosis. Furthermore, we calculated the cumulative distance between self-reporting and the diagnosis. When analyzing the similarity between the different data modality curves, it should be borne in mind that the diagnosed values were collected every three weeks with a few telephone interviews in between. They therefore do not capture fluctuations at intervals of a few days. Thus neither the results of the self-assessment nor those of the sensor recognition will match the ground truth perfectly. The main interest therefore lies in analyzing the similarity (shape and progression) of the curves. This was done using the Dynamic Time Warping DTW technique [10]. DTW is a technique to find an optimal alignment between two given (time-dependent) sequences, independent of the number of data points

Patients	selfAss	Sensors	decrease	%
p1	26.15	0.79	25.36	-96.96
p2	0.33	0.05	0.27	-83.86
p3	0.29	0.62	-0.32	109.50
p4	8.69	0.29	8.40	-96.65
p5	21.60	0.60	21.00	-97.21
p6	6.27	0.54	5.73	-91.40
p7	5.10	5.09	0.01	-0.18
p8	12.76	0.02	12.75	-99.88
p9	2.48	0.01	2.47	-99.49
p10	0.62	0.01	0.61	-98.38
mean	8.43	0.80	7.63	-65.45

Table 2: Cumulative distance between diagnosis and the self-reports (SelfAss) and between diagnosis and sensor recognition results (Sensors)

within the two different sequences. To keep the comparison fair (the self-assessment values are not necessarily in the same range as the sensor values), we normalized all three curves between 1 and 0 before calculating the minimum distance between them.

Table 3 shows the results. For example, for p1 the cumulative distance between self-reporting (SelfAss) and the diagnosis is 26.15, between sensors and diagnosis it is only 0.79. This means the distance of the sensors to the diagnosis is over 95% smaller (and thus more similar) than the distance of the self-reporting to the diagnosis. In total, for 8 out of 10 patients, the similarity of the sensor results is over 80% better than the similarity of the self-reporting. In only two patients the sensor results are not distinctly better than the self-reporting.

4 DEFERRED SELF-PERCEPTION

As stated in the introduction, psychiatrists report that the patient's perception of change can be delayed for several days. Until today, such impressions have been impressions only, and have not been evaluated objectively or in detail. With data available in different modalities (self-reporting, ground truth, sensor data), we attempted such an objective analysis. Since both kinds of data (ground truth and self-reports) were available as curves, the idea was to evaluate a potential delay in self-perception by moving (shifting) the ground truth day by day along the timeline of the self-report and calculating with every day's shift, how well the two curves match. To calculate the similarity between the curves we used the dynamic time warping (DTW) technique ([10] to find the best match between the curves, and then calculated the cumulative distance (CD) as a measure of similarity (the smaller the CD the better the match). In other words, we calculated the similarity (DTW and CD) between the two curves, then shifted the ground truth curve by one day and calculated the similarity again. As the psychiatrist estimated the deferral of self-perception of up to 7 days, we performed this shift and calculated the similarity 7 times.

Table 3 shows the results of this evaluation and includes the cumulative distance between the diagnoses state curve and the shifted self-assessment (shifted by 1-7 days) for all 10 patients. The cumulative distance is lowest for most patients (and therefore the similarity best) with an average shift of around 5 days (median

Patients	p1	p2	p3	p4	P5	p6	P7	p8	p9	p10	Average
Change Detection recall	91	86	97	100	97	100	96	95	100	100	96.5
Change Detection precision	93	96	92	94	97	87	97	95	97	91	94.2
State Recognition accuracy	70	84	68	82	71	77	74	79	83	68	76

Table 1: Summary of results of the sensor evaluation based on mobility and activity data: detection of state change Recall and Precision, and accuracy of recognition of particular states (more details were published in [3] and [2]), Numbers are in %

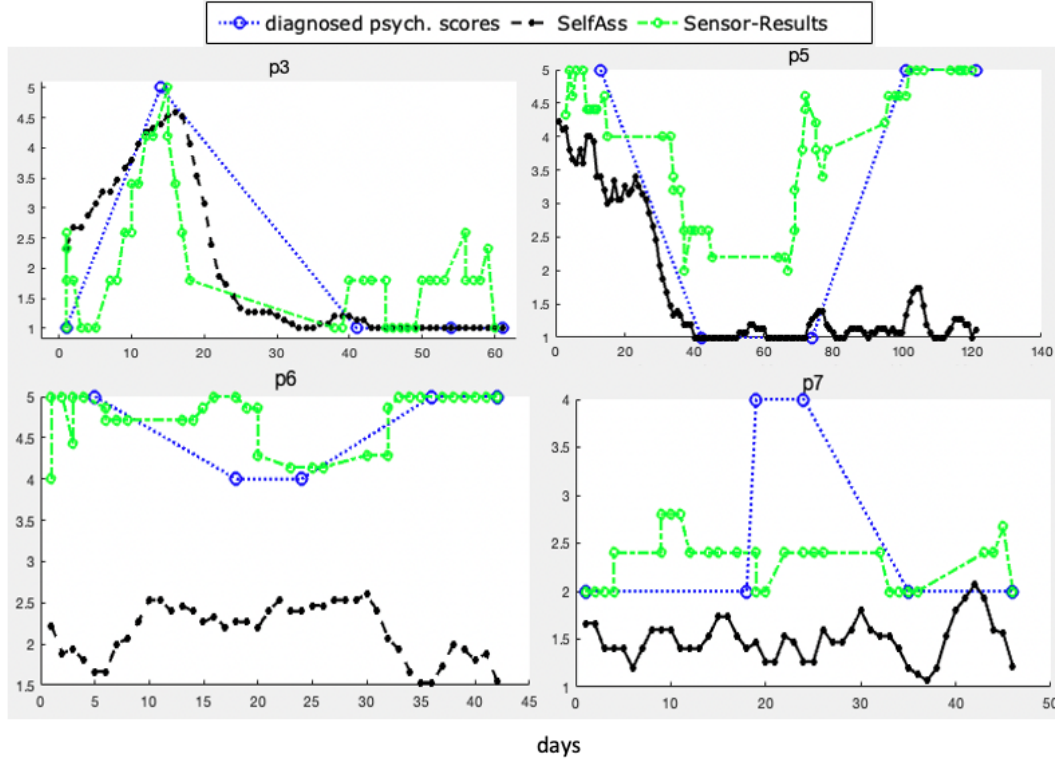


Figure 1: Visual comparison of diagnosed psychiatric scores, self-reporting, and sensor-based state detection of 4 different patients.

5, average 4.2, mode 6 - mode meaning the most often occurring best shift, which in this case is the most informative measure). This clearly supports the correctness of the psychiatrists' observation of a deferred perception of change for most patients! Nevertheless, it is worth looking into the results with more detail, because the shift varies quite a lot between patients. 3 Patients (p3, p4, and p5) have a high similarity of self-assessment and diagnosis with less than 3 days shift. For 6 (out of these 10) patients, the distance is lowest with a shift of more than 4 days. For almost all patients a shift of 4 days is not best at all!

Therefore, it would be difficult to simply apply a 4-day average shift to the self-assessment in order to enhance the accuracy. Furthermore, the current results provide no information on whether the shift is consistent or might change with time. These results support the psychiatrist's subjective observation and the hypothesis of this evaluation that a delay in change perception is measurable. Nevertheless, this analysis also shows the uniqueness of the perception delay for each person.

5 DISCUSSION AND CONCLUSION

The goal of this work was to provide a better understanding of the potential of sensor-based monitoring of mental state in comparison to self-reporting and human assessments with their drawbacks. This was done by evaluating the self-reporting of patients suffering from mental disorders and comparing it against the actual diagnosis retrieved from standard psychological scale tests but also with a sensor-based detection of mental state retrieved from the patients' smartphones.

A qualitative visual comparison of these three modalities in terms of mental state immediately gives an impression of the available data and highlights some limitations of self-reporting. In the graphs, it becomes obvious that for some patients the self-reporting does not match the diagnosis, not even close. It also provides hints that standard diagnosis is not always done at the right time.

A direct statistical comparison of self-reporting and diagnosis, furthermore reveals a delay of self-perception for many patients. Data of most of them (60% of the patients) shows a shift of 4 or more days, and only 20% seem to be able to report state changes correctly within 1 day. In regards to effective treatment of depressive or manic

shift	p1	p2	p3	p4	p5	p6	p7	p8	p9	p10	average/median/mode
1	4.2	2.3	0.0	0.5	0.0	5.1	0.6	3.8	0.7	2.8	
2	3.4	1.4	0.3	0.2	1.2	6.5	2.8	8.9	0.9	7.4	
3	2.8	4.6	1.9	2.4	5.4	3.9	1.1	4.2	0.7	10.5	
4	4.4	2.9	1.5	1.2	0.9	4.4	1.2	6.1	3.6	1.4	
5	6.2	1.2	2.0	3.8	3.1	5.0	0.4	2.9	0.1	7.6	
6	3.1	3.2	0.0	0.8	2.1	0.4	0.0	2.6	0.3	5.7	
7	9.6	1.5	0.0	2.1	2.4	5.7	0.1	3.2	0.2	0.6	
best	3	5	1	2	1	6	6	6	5	7	4.2 / 5 / 6

Table 3: Cumulative distance between ground truth (diagnosis) and self-reporting (with shifted diagnosis by 1-7) days.

episodes, the timely detection of the onset of change is crucial. Thus, a deferred perception of a starting change, and possibly even more delay until a person acknowledges the change and seeks help can lead to a manifested episode. This in turn requires more severe treatment (heavier doses of medication) and longer-lasting episodes than it would have been with immediate adjustments. This aspect, in particular, highlights the potential of objective sensor-based monitoring of behavior patterns since it would help detect the onset of change right away, and also shows the downside of mainly relying on the human ability of self-assessing.

A closer look at patient p3 reveals some interesting aspects, but also some limitations of this work. Patient 3 is the only patient whose self-assessment seems to be closer to the actual condition than the recognition results. On the one hand, there are patients whose self-perception is quite accurate. On the other hand, the absolute values of the cumulative distances show that the difference between self-assessment and recognition results is relatively small. With a distance of 0.29 and 0.62 (difference -0.33) compared to patient 1, for example (26.15 and 0.79), both are very similar to the diagnosed condition. In addition, a look at the graphs of p3 shows the limitations resulting from the fact that a baseline value is only available every three weeks. Looking at the peak around day 15, the curves show that the sensors detect the decline in the condition a few days earlier than the self-assessment. It is difficult to say which of the curves is the correct one, as the psychiatric score curve does not include the values of these specific days. Given the assessment of the shift in patient perception and the results of the similarity analysis in the previous chapter, the interpretation that the detection is probably more accurate than the self-assessment for this patient as well is supported.

Closing the discussion we want to highlight, that the state recognition and self-assessment analysis is based on a real-life data set, that was recorded during the typical daily lives of bipolar patients. During the evaluation, we had to deal with data loss due to switched off or not charged smartphones. Even though the number of patients is not large, the results presented here are valid in a real-life setting, which emphasizes the value and relevance of these results. In the treatment of mental health disorders, contact with other people is essential, and for patients with affective mood disorders, frequent self-assessments also serve to help them understand their conditions. Thus we want to stress as well, that our analysis does not intend to replace humane interaction and self-assessments, but to show the potential of sensor-based monitoring as an addition for

both, patient and physician to optimize the management of mental disorders and with this help to reduce negative effects for them.

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